AP PRECALCULUS

Unit 1 • Practice

Polynomial and Rational Functions

AP-style practice • Multiple choice and free response

How to use this practice set

The questions follow the format of the AP Precalculus exam. **Part A**: multiple choice, no calculator. **Part B**: multiple choice, graphing calculator allowed. **Part C**: free response; show your work and justify your answers with words, as on the exam.
Suggested timing: Part A 30 min • Part B 10 min • Part C 45 min.

Part A • Multiple choice • No calculator

Question 1 (1.1)

The table gives values of a function f at selected values of x.

x	0	2	4	6
$f(x)$	5	9	11	12

Which statement is consistent with the table?

(A) f is increasing and its graph is concave up.

(B) f is increasing and its graph is concave down.

(C) f is decreasing and its graph is concave up.

(D) f is decreasing and its graph is concave down.

Question 2 (1.2)

What is the average rate of change of $f(x) = x^3 - 2x$ on the interval $[-1, 2]$?

(A) 3

(B) $\frac{5}{3}$

(C) 1

(D) -1

Question 3 (1.3)

x	1	2	3	4	5
y	2	5	10	17	26

Which type of function best models the data in the table?

(A) Quadratic

(B) Linear

(C) Cubic

(D) Constant

Question 4 (1.5)

A polynomial p of degree 4 has real coefficients. Two of its zeros are 2 and $1 + 3i$. Which of the following must also be a zero of p?

(A) -2

(B) $3 + i$

(C) $-1 - 3i$

(D) $1 - 3i$

Question 5 (1.6)

Let $p(x) = x^2(x-3)^3(x+1)$. What is $\lim_{x \to -\infty} p(x)$?

(A) $-\infty$ (B) $+\infty$
(C) 0 (D) 1

Question 6 (1.5)

Which statement describes the graph of $p(x) = (x+2)^2(x-1)$ at $x = -2$?

(A) It has a vertical asymptote. (B) It crosses the x-axis.
(C) It touches the x-axis without crossing it. (D) It has a hole.

Question 7 (1.7)

Which line is a horizontal asymptote of the graph of $r(x) = \dfrac{3x^2 - x}{6x^2 + 5}$?

(A) $y = 3$ (B) $y = 0$
(C) $y = -\dfrac{1}{5}$ (D) $y = \dfrac{1}{2}$

Question 8 (1.10)

Let $r(x) = \dfrac{x^2 - 1}{x^2 + x - 2}$. Which statement is true about the graph of r at $x = 1$?

(A) It has a vertical asymptote $x = 1$. (B) It has a hole at $\left(1, \frac{2}{3}\right)$.
(C) It has a zero at $x = 1$. (D) It has a hole at $(1, 0)$.

Question 9 (1.9)

What is $\lim_{x \to 2^+} \dfrac{x+1}{2-x}$?

(A) 0 (B) 3
(C) $+\infty$ (D) $-\infty$

Question 10 (1.11)

The graph of $r(x) = \dfrac{2x^2 + 3x - 4}{x + 2}$ has a slant asymptote. What is its equation?

(A) $y = 2x - 1$ (B) $y = 2x + 7$
(C) $y = 2x + 3$ (D) $y = x - 1$

Question 11 (1.11)

What is the coefficient of x^2 in the expansion of $(x-3)^4$?

(A) 6 (B) -108
(C) 54 (D) 9

Question 12 (1.12)

The function f has domain $[0, 8]$ and range $[-2, 3]$. What is the range of $g(x) = -2f\left(\frac{x}{2}\right) + 1$?

(A) $[-5, 5]$ (B) $[-3, 7]$
(C) $[-4, 6]$ (D) $[-1, 3]$

Question 13 (1.5)

Which function is even?

(A) $f(x) = x^3 - x$
(B) $f(x) = x^4 + 3x^2 - 1$
(C) $f(x) = x^2 + x$
(D) $f(x) = (x-1)^2$

Question 14 (1.8)

What is the solution set of $\dfrac{x-3}{x+1} \geq 0$?

(A) $(-\infty, -1] \cup [3, \infty)$
(B) $[3, \infty)$
(C) $(-1, 3]$
(D) $(-\infty, -1) \cup [3, \infty)$

Part B • Multiple choice • Graphing calculator

Question 15 (1.14)

An open box is made from a 16 in by 10 in sheet by cutting squares of side x in from the corners. Its volume is $V(x) = x(16-2x)(10-2x)$ for $0 < x < 5$. What is the maximum volume of the box?

(A) 120 in^3
(B) 144 in^3
(C) 160 in^3
(D) 168 in^3

Question 16 (1.14)

The height h, in meters, of a model rocket was measured at times t, in seconds.

t	0	1	2	3	4
$h(t)$	1.2	6.1	9.0	9.9	8.6

Using a quadratic regression, what is the predicted height at $t = 5$ seconds?

(A) 4.2 m
(B) 5.3 m
(C) 6.4 m
(D) 7.3 m

Part C • Free response

Question 17 (Function concepts • no calculator)

The polynomial function h has degree 3 and leading coefficient 1. Selected values of h are given in the table.

x	-2	-1	0	1	2	3
$h(x)$	-1	3	1	-1	3	19

(A) (i) Find the average rate of change of h over the interval $[-1, 1]$. Show the work that leads to your answer.
(ii) How many real zeros does h have? Give a reason for your answer using the table.
(B) (i) The average rates of change of h over $[1, 2]$ and $[2, 3]$ are different. Use them to determine whether the graph of h is concave up or concave down on $[1, 3]$. Give a reason for your answer.
(ii) Write a limit statement that describes the end behavior of h as x decreases without bound.
(C) Let $k(x) = h(x-2) + 4$. Find $k(3)$, and describe how the graph of k is obtained from the graph of h.

Question 18 (Modeling a non-periodic context • calculator)

A theater studies how its revenue R, in dollars, depends on the ticket price p, in dollars.

p (dollars)	10	15	20	25	30
R (dollars)	4000	5250	6000	6250	6000

(A) (i) Explain why a quadratic function is an appropriate model for these data.
(ii) Write an expression for a quadratic function $R(p)$ that fits the data exactly.
(B) (i) Use your model to predict the revenue when the ticket price is \$35.
(ii) Find the average rate of change of R from $p = 25$ to $p = 35$, and interpret its meaning in context, with units.
(C) Revenue cannot be negative. Find the interval of ticket prices for which the model gives a revenue of at least \$0, and state one reason the model may not be valid for every price in that interval.

Question 19 (Symbolic manipulations • no calculator)

Let $r(x) = \dfrac{2x^2 + 2x - 4}{x^2 - 1}$ and $p(x) = x^3 - 2x^2 - 5x + 6$.

(A) (i) Find the zeros of r and the coordinates of any hole in the graph of r.
(ii) Find the vertical asymptote of the graph of r. Write limit statements that justify your answer.
(B) (i) Write a limit statement that describes the end behavior of r.
(ii) Solve $r(x) \geq 0$. Express your answer in interval notation.
(C) It is known that $p(1) = 0$. Write $p(x)$ as a product of linear factors, and find all zeros of p.

Check your answers with the answer key for this unit.